BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Paper Chemistry - 11th — Annual Exam – 2021 — SECTION- A

Time Allowed: 20 mints | **Maximum Marks:** 18

Note: Q.1: Fill in the blanks with sutable option.

Q1. How many moles are in 6.023 x 10^{23} fromula units of MgO?

(A) 0.5 moles
(B) 1.5 moles
(C) 1 mole
(D) 2 moles

Q2. Bohr model of Hydrogen atom is based on.....

(A) Ruther ford model
(B) Dalton's model
(C) Plank's model
(D) Einstein theory

Q3. Which one of the following will have angular geometry?

(A) NH_3
(B) $SnCl_2$
(C) $Becl_2$
(D) BF_3

Q4. The S.I unit of pressure is.....

(A) Atm
(B) psi
(C) Nm^{-2}
(D) all of these

Q5. The attractive forces within a molecule of a liquid are called.....

(A) Intermolecular forces
(B) hydrogen bonding
(C) Intra molecular forces
(D) vander Waal forces

Q6. Which one is the example of liquid cyrstals?

(A) Benzo cane
(B) Stearin
(C) Hydrazine
(D) None

Q7. A cubic crystal has .....centres of symmetry.

(A) One
(B) two
(C) three
(D) four

Q8. The value of Kc is independence of the.....

(A) Temperature
(B) Initial concentration
(C) Pressure
(D) None of these

Q9. 10-3 moles of HNO3 is dissolved /liter, its pH is .....

(A) -3
(B) 5
(C) 3
(D) 1

Q10. No work is done at constant.....

(A) Pressure
(B) Volume
(C) Temperature
(D) mass

Q11. The fuel used in cell can be .....

(A) Gas
(B) liquid
(C) solid
(D) all of these

Q12. ΔH is positive when heat is .....

(A) Obsorb ed
(B) released
(C) removed
(D) none

Q13. Which one is not electromagnetic?

(A) IR rays
(B) Uv rays
(C) γ rays
(D) cathode rays

Q14. The efficiency of chemical reaction can be determind from .....

(A) Actual yield
(B) theoretical yield
(C) percentage yield
(D) quantum yield

Q15. Weak acid and its salt with a strong base will have pH?

(A) More than 7
(B) less than 7
(C) neutral
(D) none

Q16. The enzyme used to convert starch into sugar during enzyme catalysis is.....

(A) Zymase
(B) Invertase
(C) ptyalin
(D) pepsin

Q17. The stability of colloidal system depends on.....

(A) Charge
(B) solvation
(C) Brownian motion
(D) all of these

Q18. Endothermic reactions are favored in forward direction by.....

(A) Cooling
(B) Freezing
(C) Heating
(D) adding catalyst

SECTION-B

Time Allowed: 2.40 Hrs. | **Maximum Marks:** 40

Q2. (Marks 40) Q.2: Attempt any ten (10) parts. All parts carry equal marks.

i. Justify the statement "Total heat provided to a system at constant volume is used to change internal energy of the system."

ii. Calculate the cell potential for the following two half cells. a) Pb^{+2}/Pb E^0_red = -0.13 V b) Cl_2/Cl^- E^0_red = + 1.36 V

iii. Calculate the percentage composition of each element is H_2SO_4.

iv. How will you differentiate between a continuous and a line spectrum?

v. Why MOT is more important than VBT?

vi. Prove that: PV = nRT

vii. HF is liquid at ordinary temperature while HCl is a gas. Why?

viii. What is the differentiate between isomorphism and polymorphism?

ix. Why a solution of $FeSO_4$ is acidic.

x. The ionization or NaCl is suppressed by passing HCl gas over it. Why.

xi. Why evaporation of a liquid causes cooling?

xii. Determine the relative retes of diffusion of H_2 and CO_2 at same condition of temperature and pressure.

xiii. Write four uses of X-rays.

SECTION-C

Maximum Marks: 27

Note: Note: Attempt any Three of the following questions. All questions carry equal marks.

Q3.

a) Define absolute temperature. How it can be represented graphically? Why it is not attainable.

b) 800 cm^3 of a gas at 400 torr pressure and 60^oC was heated up to 832 k. What is final volume of the gas?

Q4.

a) Prove that: $K_p = K_c = K_x$

b) Calculate Kc for synthesis of ammonia at 500^oC where Kp = 1.44 x 10^{-5}

Q5.

a) Enlist the postulates of collision theory for chemical reaction.

b) Define Molarity? When 4.5 g of glucose dissolved in 100 g of water, calculate its molarity. Molecular mass of glucose = 180g. mol^{-1}.

Q6.

a) What is enthalpy of a system? Prove that $\Delta H = q_p$

b) xiv. How can we measure Electrode Potential of SHE?